Neel Gali

Software Engineer

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | AI-Powered Digital Pathology Platform

- Engineered a **high-performance whole-slide image viewer** using **React** and **TypeScript** for **billion-pixel microscopy images**, improving scientists' analysis speed by 40%.
- Developed **ML annotation pipelines** using **Python** and **TensorFlow** to auto-analyze **tissue response** on large datasets, reducing manual annotation time by 65%.
- Designed and implemented robust **tissue specimen tracking systems** with **Ruby on Rails** for histology workflow management, ensuring 99.9% data integrity across 5000+ samples.
- Built advanced **annotation and analysis tools** for pathologists using **React** and **D3.js**, facilitating precise scoring of **pathology findings** and regions of interest.
- Integrated **image alignment and overlay** features for **serial sections** using **OpenCV** and **JavaScript**, enhancing comparative analysis capabilities by 30% for users.
- Led the deployment of **model predictions** interface into the **whole-slide image viewer**, enabling scientist **validation** and accelerating diagnostic throughput by 25%.
- Architected a scalable microservices platform using **AWS ECS** and **Kubernetes** to support growing demands of **digital imaging** and data processing, reducing latency by 20%.

Software Engineer *Jul 2023 – Jun 2025*

Samay Systems LLC | Enterprise Data Analytics Platform

- Developed scalable **data pipelines** using **Python** and **Apache Spark** on **AWS S3** for ingesting 10TB+ daily enterprise data, improving processing efficiency by 30%.
- Implemented **RESTful APIs** with **Java Spring Boot** to expose analytics data to client applications, achieving 99.9% uptime and supporting 1000+ concurrent requests.
- Designed and optimized **PostgreSQL** databases for high-volume transactions, resulting in query performance improvements of 45% and reduced reporting generation times.
- Created interactive dashboards using **React** and **D3.js** for data visualization, providing key business insights to 200+ internal stakeholders monthly.
- Automated data validation and monitoring scripts using **Python** and **Airflow**, reducing manual error detection time by 70% and ensuring data quality.
- Contributed to CI/CD pipeline implementation with **Jenkins** and **Docker**, streamlining deployment processes and accelerating feature delivery cycles by 25%.
- Leveraged **AWS Lambda** and **SQS** to build event-driven microservices, improving system responsiveness and reducing operational costs for a critical module by 18%.
- Developed a user authentication and authorization module using **JWT** and **OAuth2**, enhancing application security and user access control for 5000+ users.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Healthcare SaaS Application Development

- Developed core modules for a healthcare management SaaS application using **Java** and **Spring MVC**, improving patient data management efficiency by 20%.
- Implemented responsive user interfaces with **HTML**, **CSS**, and **JavaScript** for physician portals, enhancing usability and reducing training time for new users.
- Designed and maintained **SQL Server** databases, optimizing schemas and queries to support high transaction volumes, improving data retrieval speed by 35%.
- Created and consumed **RESTful APIs** to integrate with third-party medical devices, enabling seamless data exchange and improving system interoperability by 15%.
- Participated in code reviews and adopted **Agile** methodologies, ensuring high code quality and on-time delivery of features in a collaborative team environment.
- Developed automated unit and integration tests using **JUnit** and **Mockito**, decreasing defect rates by 25% and improving software reliability.

Skills

Concepts: Microservices, Agile, CI/CD, RESTful APIs, Event-driven Architecture, JWT, OAuth2, Data Pipelines, Digital Imaging, Computer Vision, Machine Learning
Languages: Python, TypeScript, JavaScript, Java, Ruby, HTML, CSS
Cloud & Infra: AWS ECS, Kubernetes, AWS S3, AWS Lambda, SQS, PostgreSQL, SQL Server
Frameworks & Tools: React, TensorFlow, D3.js, OpenCV, Apache Spark, Spring Boot, Airflow, Jenkins, Docker, Spring MVC, JUnit, Mockito, Ruby on Rails

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India